

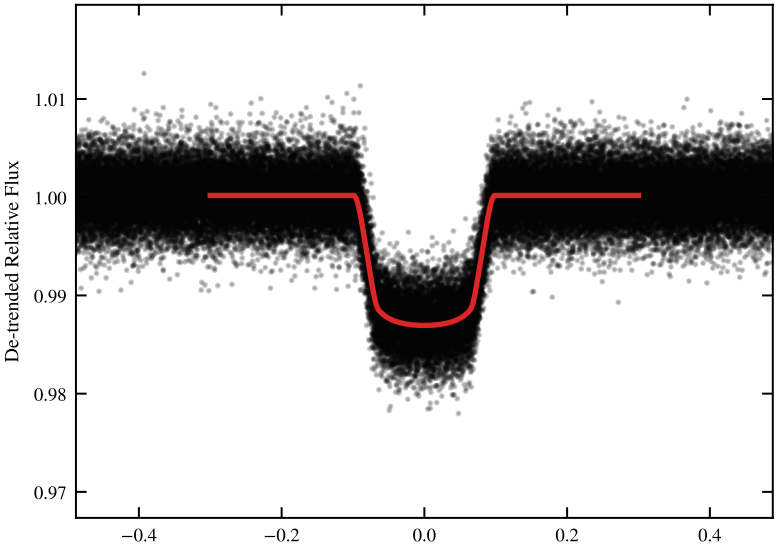
TESS PLANET TRANSIT CHARACTERIZATION REPORT

Target: TIC 381979901 | RA: 77.60540° | Dec: -59.53566° | Sectors: 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39

Derived Planetary Parameters

Parameter	Derived Value (MCMC)
Period (d)	3.2541 ± 3.58e-06
Rp/R★	0.11212 ± 0.000456
Rp (R⊕)	15.623 ± 0.0635
b	0.63851 ± 0.0117
T14 (h)	4.6868 ± 0.0136
a (AU)	0.029079 ± 0.00028
Teq (K)	1949.4 ± 33.1

Primary Phased Transit



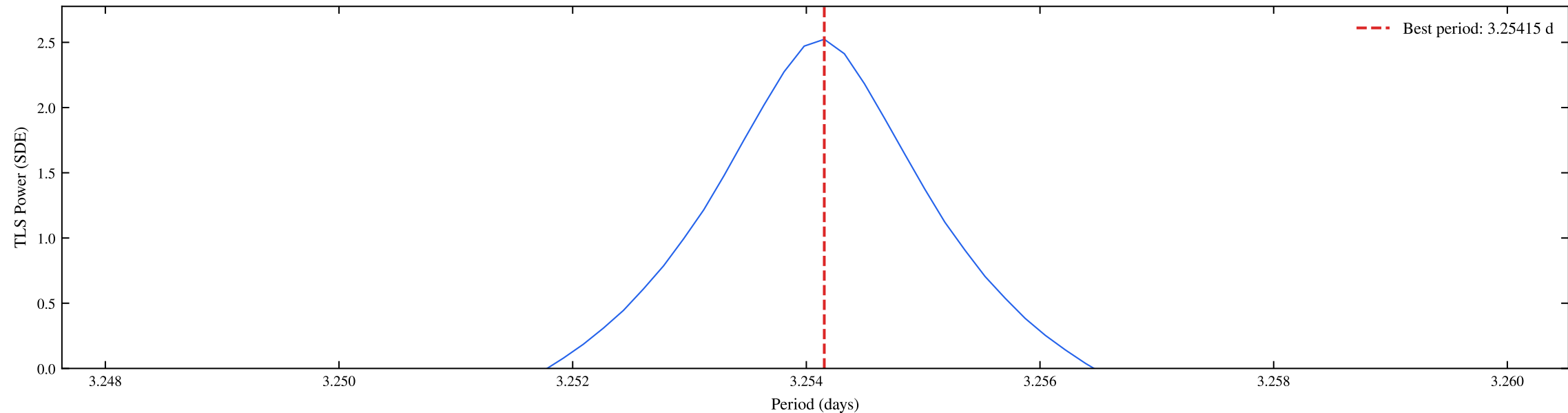
Literature Comparison & MCMC Diagnostics

Parameter / Metric	Literature Value	Derived / Diagnostic
P1 Period (d)	3.25415	3.254121 ± 0.00000
P1 Epoch (BTJD)	N/A	2038.9068 ± 0.0002
Max R-hat (R)	N/A	1.0012
Min ESS	N/A	4800
MCMC Divergences	N/A	0

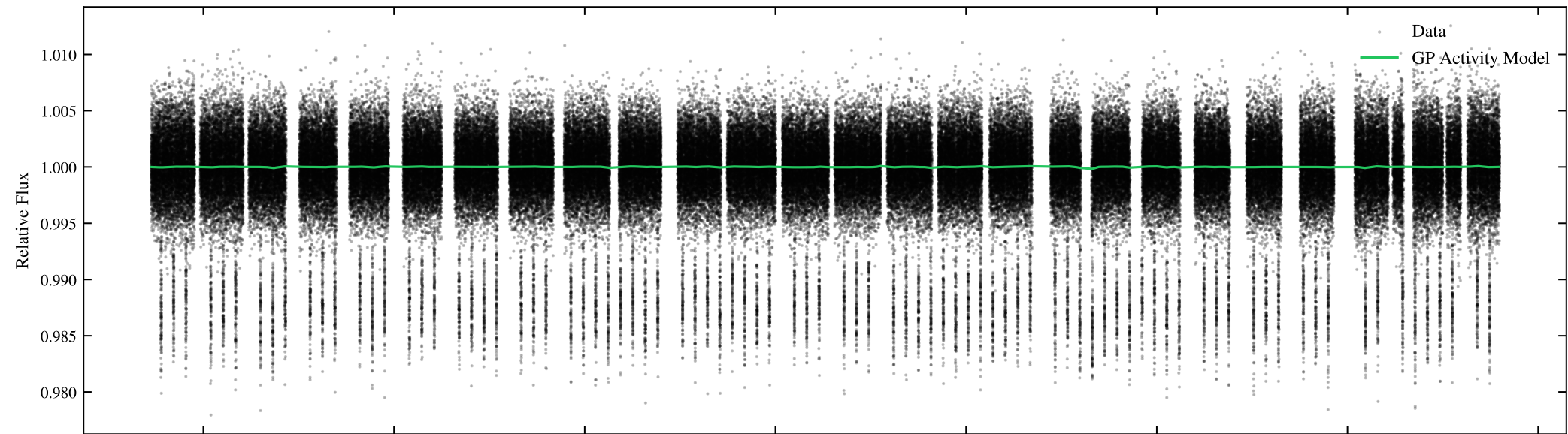
Host Star Properties

Stellar Parameter	Value	Source
R★ (R☉)	1.276	VizieR TIC8
M★ (M☉)	1.112	Torres 2010
Teff (K)	6102	VizieR TIC8
log g (dex)	4.286 ± 2	Gaia DR3
[Fe/H]	N/A	Catalog
ρ★ (g/cm³)	0.7548	Derived

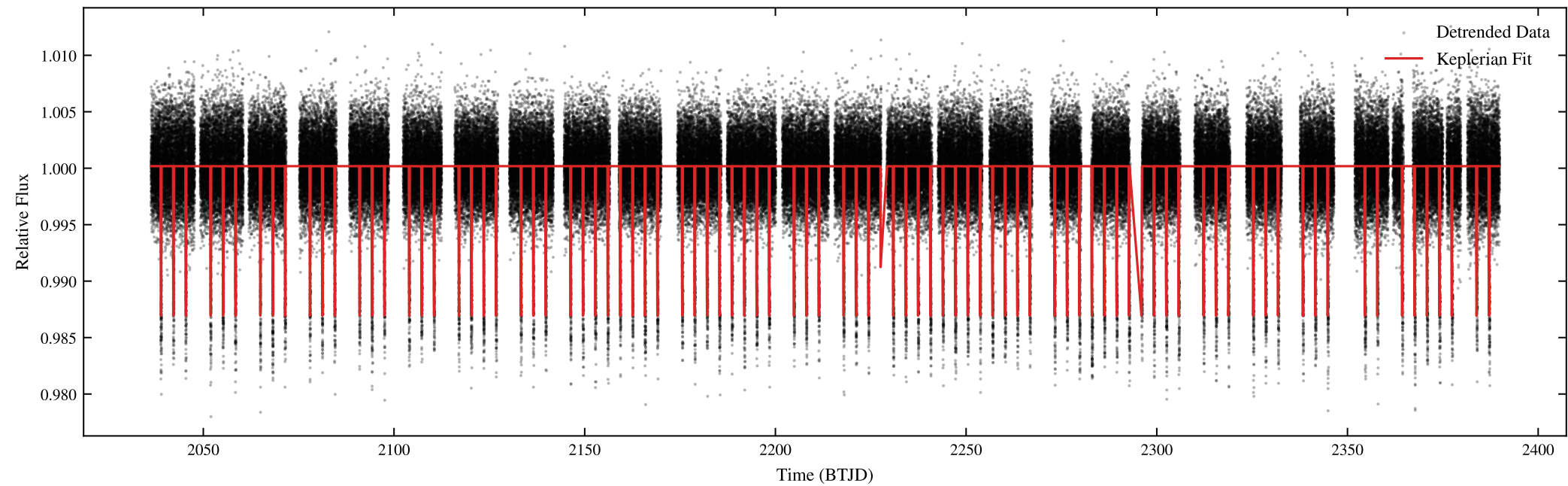
TIC 381979901 | Sectors: 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39 | TLS Periodogram (SDE = 2.52)



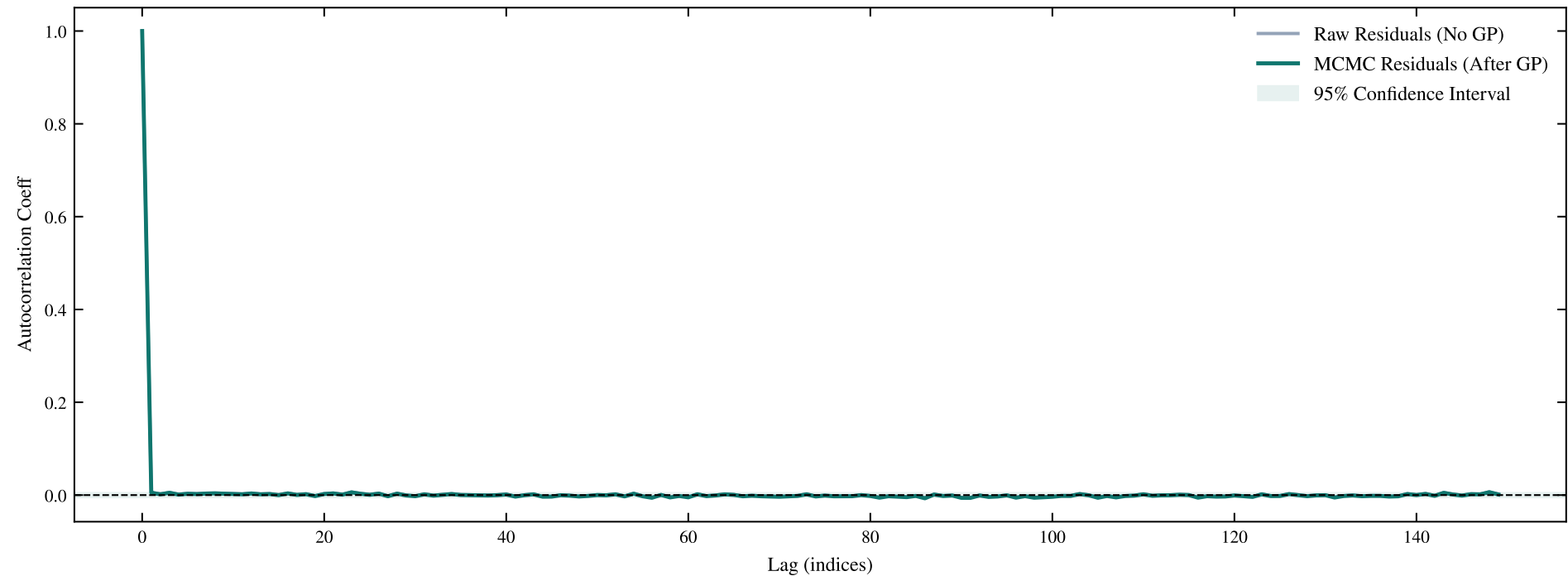
TIC 381979901 | Sectors: 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39 | Bayesian GP Detrending (Time Domain)



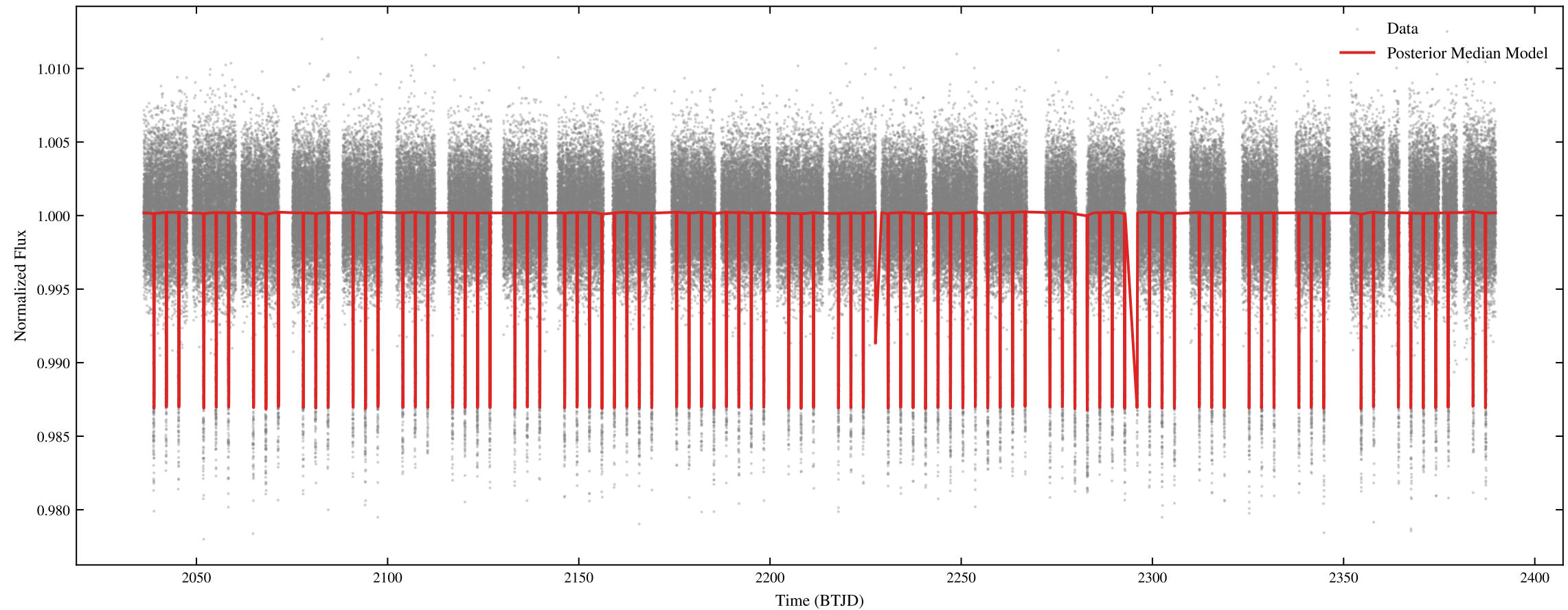
Detrended Light Curve & Best-Fit Transit Model

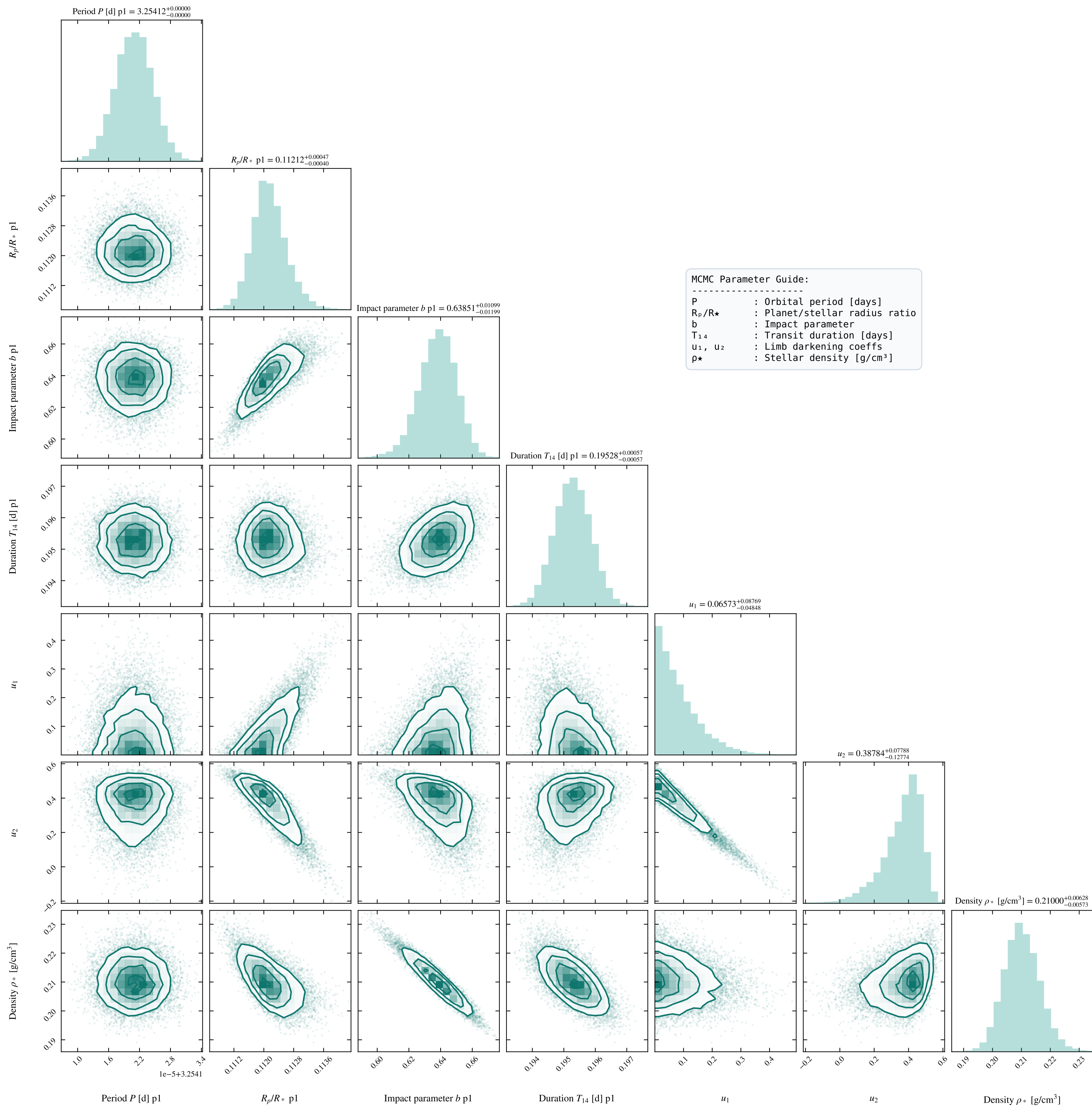


TIC 381979901 | Sectors: 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39 | GP Residuals Autocorrelation Function (ACF)

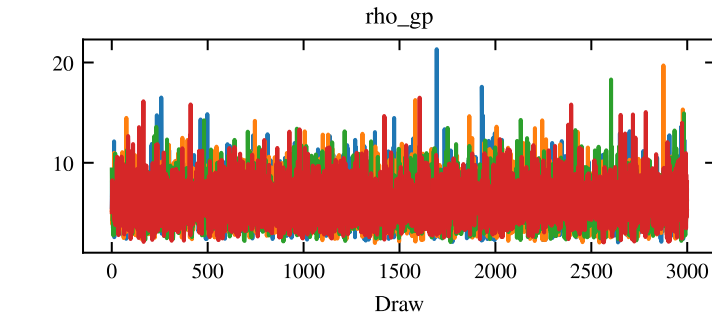
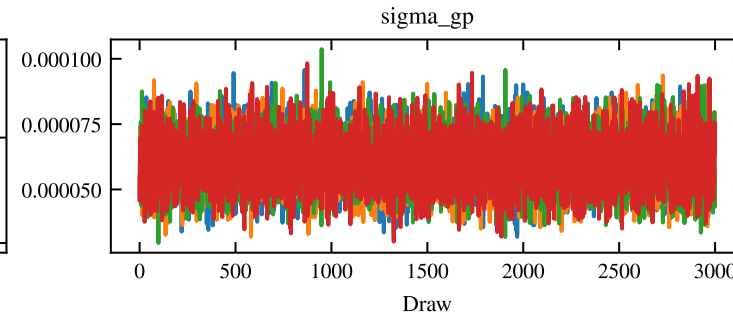
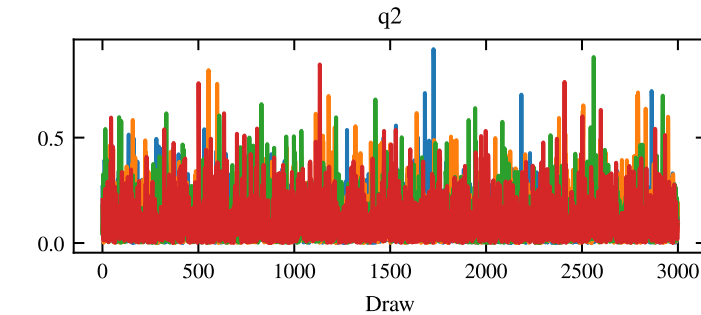
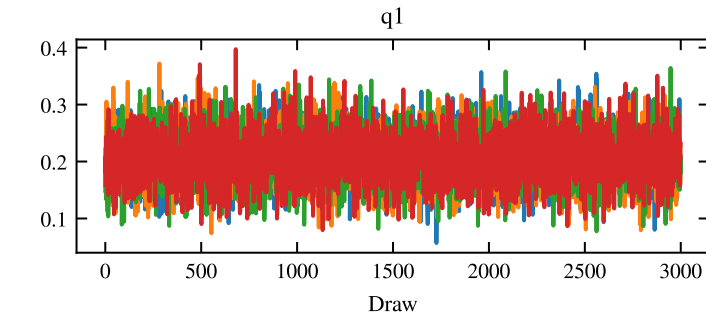
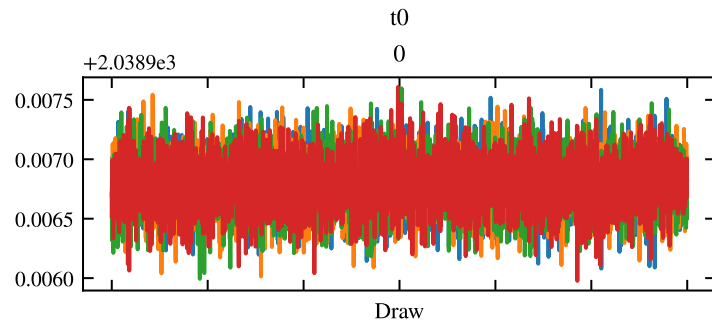
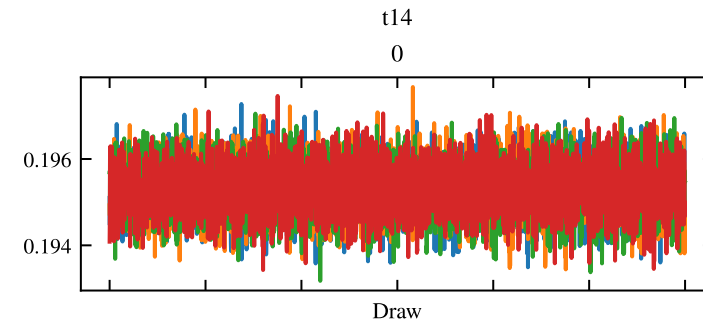
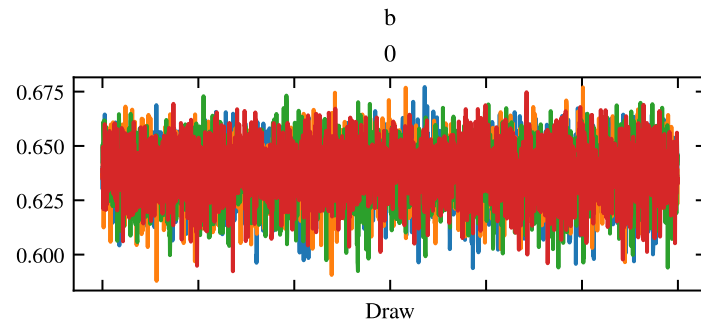
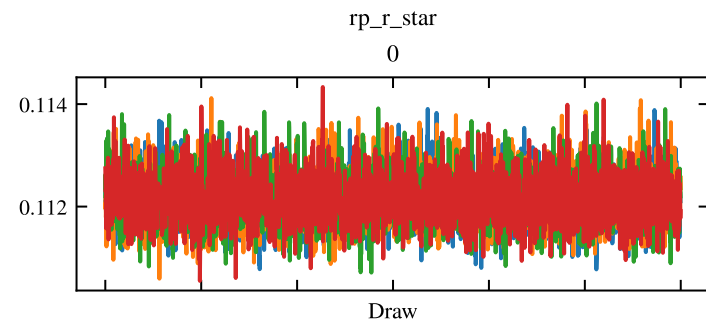


TIC 381979901 | Sectors: 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39 | MCMC Posterior Predictive Check

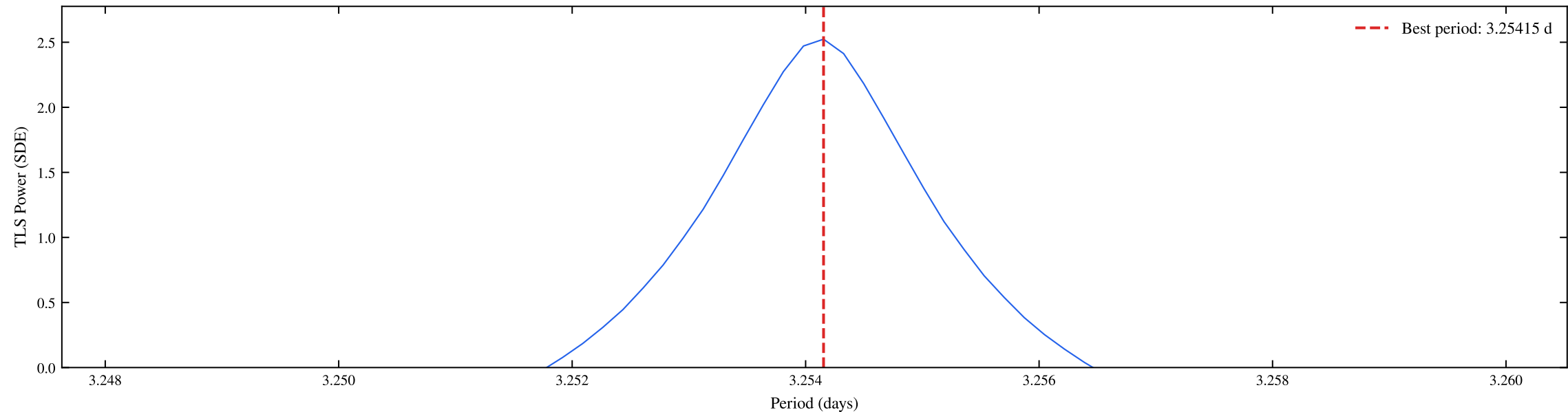




TIC 381979901 | Sectors: 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39 | MCMC Trace Plots



TIC 381979901 | Sectors: 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39 | TLS Coarse Periodogram (SDE = 2.52)



TIC 381979901 | Sectors: 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39 | TLS Fine Periodogram (SDE = 2.52)

